

Introduction to lattice based cryptography

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September 2026, Barcelona

Motivation

- till now most cryptography relied on hardness of finding prime factors or finding the discrete logarithm, ie. RSA, Elliptic Curves, etc
- Shor's algorithm suggests that factorization can be done in polynomial time (on quantum computers).
 - given an odd composite $n \in \mathbb{Z}$, find $d \in \mathbb{Z}$, such that $1 < d < n$ and $d|n$.
- even if quantum computers never reach, the threat is there, so we must prepare
- since 2023/2024 Chrome and Firefox have lattice-based cryptography to the browser for TLS
 - hybrid: X25519 + ML-KEM to establish a symmetric encryption key
 - protect against *harvest now, decrypt later* attacks

Motivation (in Ethereum's ecosystem)

- Ethereum, is going with hash-based signatures for the validators.
 - aggregation through SNARK proofs (eg. binus, flock, etc)
 - each slot a different XMSS leaf
 - the most conservative choice in terms of security
- hash-based simpler to implement, less potential errors.
Whereas lattice-based have more subtleties
- hash-based schemes don't have much algebraic structure
 - good for less attacks, not so fun to build stuff with
 - eg. elliptic curves have algebraic structure, allows for signature aggregation, thresholds, etc
- lattice-based schemes have smaller proofs, signatures and public keys
 - better suited for end-user wallets

Post Quantum Cryptography - Status of the ecosystem

- isogeny based
- hash based
- codes based
- lattice based
- multivariate based

TODO

Preliminaries

Recap of concepts:

- Rings
 - ideals
- Vector Spaces
- Modules
- Kernel

Rings

Definition (Ring)

A set A with operations called *addition* and *multiplication* which satisfy the following axioms:

- i. A with addition alone is an abelian group.
- ii. Multiplication is associative.
- iii. Multiplication is distributive over addition. That is, $\forall a, b, c \in A$,

$$a(b + c) = ab + ac$$

$$(b + c)a = ba + ca$$

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Eg: Ring $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, 3, \dots\}$, Ideal $n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$ for fixed $n \in \mathbb{Z}$.

Vector spaces

Definition (Vector space)

A *vector space* over a field F is a set V , with two operations $+$, $\cdot$, called *vector addition* and *scalar multiplication*, such that

- V with vector addition is an abelian group
- $\forall k \in F$ and $\vec{a} \in V$, the scalar product $k\vec{a}$ is an element of V , subject to the following conditions: $\forall k, l \in F, \vec{a}, \vec{b} \in V$
 - i. $k(\vec{a} + \vec{b}) = k\vec{a} + k\vec{b}$
 - ii. $(k + l)\vec{a} = k\vec{a} + l\vec{a}$
 - iii. $k(l\vec{a}) = (kl)\vec{a}$
 - iv. $1\vec{a} = \vec{a}$
- *basis*: set of vectors that *span* the vector space

Modules

Definition (Module)

Let A be a ring. An A -module is an Abelian group M with a multiplication map

$$\begin{aligned} A \times M &\longrightarrow M \\ (f, m) &\longmapsto fm \end{aligned}$$

satisfying $\forall f, g \in A, \quad m, n \in M$.

- i. $f(m \pm n) = fm \pm fn$
- ii. $(f \pm g)m = fm \pm gm$
- iii. $(fg)m = f(gm)$
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Remark: Vector spaces are defined over fields, a Module is the generalization of a vector space where the scalars come from a ring instead of a field.

Kernel

TODO

Intuition: set of inputs that a function completely "erases", ie. they all get mapped to zero.

Lattices

Definition (Lattice)

A lattice $\mathcal{L}$ in $\mathbb{R}^n$ is the set of all integer linear combinations of m linearly independent vectors $b_1, \dots, b_m \in \mathbb{R}^n$, with $m \leq n$.

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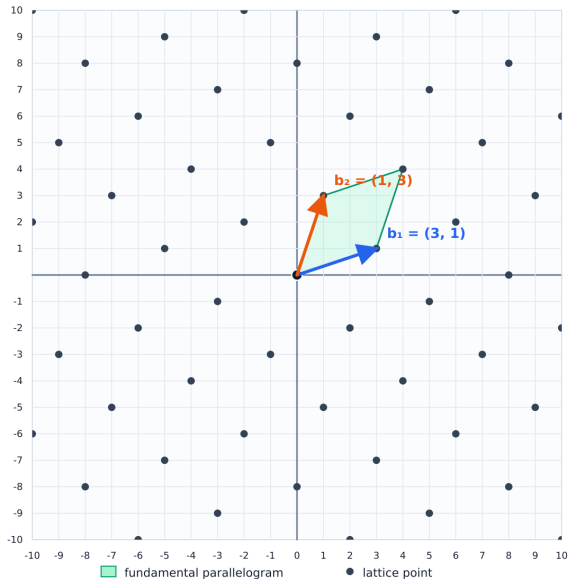
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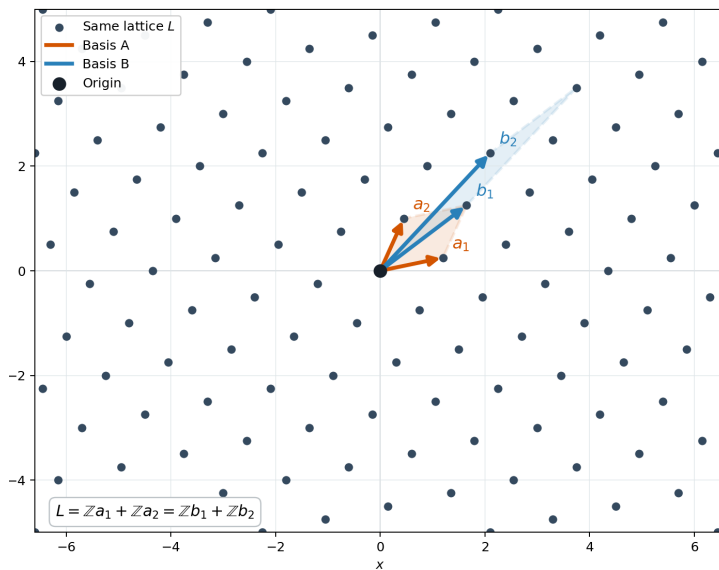
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In most cryptographic applications, $b_1, \dots, b_n \in \mathbb{Z}^n$, ie. $\mathcal{L}$ is an integer lattice.

2-dimensional lattice

$$\Lambda = \{n b_1 + m b_2 : n, m \in \mathbb{Z}\}, b_1 = (3, 1), b_2 = (1, 3)$$





Lattice properties

a n -dimensional lattice $\mathcal{L}$ is

- i. *additive subgroup* of $\mathbb{R}^n$: $\mathcal{L}$ is the nonempty subset of $\mathbb{R}^n$ closed under vector addition (and subtraction),

$$\text{ie. } \forall x, y \in \mathcal{L}, -x, x + y \in \mathcal{L}$$

- ii. *discrete*: every $x \in \mathcal{L}$ has a neighborhood in $\mathbb{R}^n$ in which x is the only lattice point.

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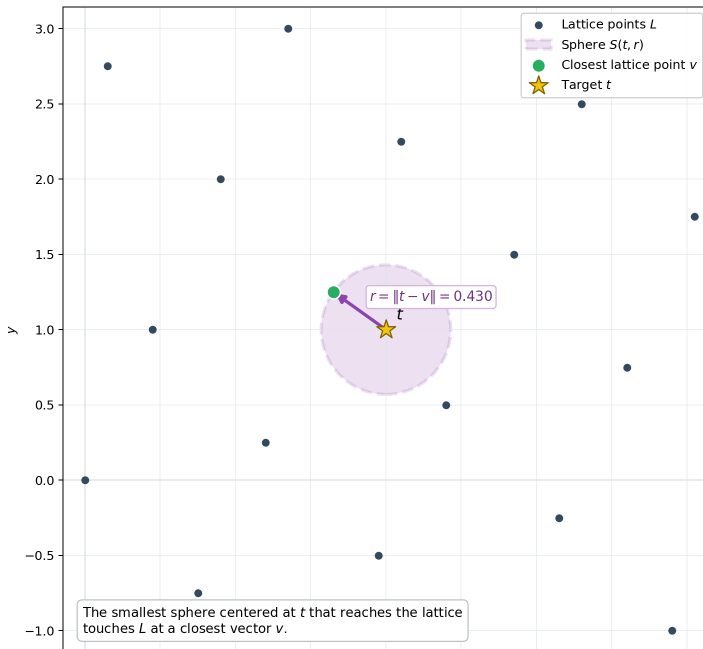
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Successive minima $\lambda_i(\mathcal{L})$: smallest radius r s. th. $\mathcal{L}$ the n -dimensional ball of radius r contains i linearly independent lattice points.

$$\lambda_1(\mathcal{L}) \leq \lambda_2(\mathcal{L}) \leq \dots \leq \lambda_n(\mathcal{L})$$

Closest Vector Problem: the target-centered sphere



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recall, a lattice: $\mathcal{L}(B) := B \cdot \mathbb{Z}^n = \left\{ \sum_{i=1}^n z_i b_i \mid z_i \in \mathbb{Z}^n \right\}$

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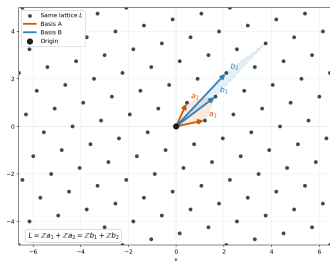
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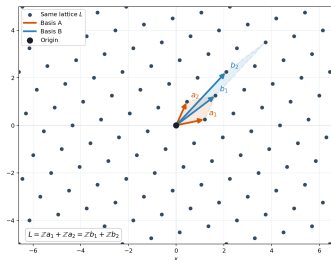
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 - the basis B is not unique
- $B \in \mathbb{Z}^{n \times n}$, n vectors, each with n coordinates $\implies$ requires $\mathcal{O}(n^2)$ integers.
 - only if we could have some better lattice structure to represent the lattice with fewer entries...



q-ary lattice

a full-rank lattice is q-ary if $q\mathbb{Z}^d \subseteq \mathcal{L} \subseteq \mathbb{Z}^d$

$\implies$ q-ary lattice is an infinite set of integer points, but membership depends only on the residue class modulo q

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q-ary lattice: $z \longmapsto Az \pmod{q}$

$$\mathcal{L}_q^\perp(A) = \ker(\psi_A) = \{z \in \mathbb{Z}^m \mid Az = 0 \pmod{q}\}$$

contains all vectors that are orthogonal modulo q to the rows of A .

Intuition: keep only the points that satisfy some modular rules. Since everything repeats every q steps, the lattice has a structured and periodic shape, convenient in cryptography.

Linear algebra modulo q , but breaking it requires solving a hard lattice problem.

Polynomial rings

- The NTRU paper (1996) introduced the usage of polynomial rings, which latter was ported over Ajtai work and other lattices, now is the general approach in Ring-SIS, Ring-LWE, Module-LWE, etc.
- Polynomial rings: represent an n -dimensional vector $(a_0, \dots, a_{n-1}) \in \mathbb{Z}_q^n$ as a single ring element (polynomial)

$$a(X) = a_0 + a_1X + \dots + a_{n-1}X^{n-1} \in \mathbb{Z}_q[X]/(X^n + 1)$$

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- Multiplication:

$$a(X) \cdot b(X) = \left(\sum_{i=0}^{n-1} \sum_{j=0}^{n-1} a_i b_j X^{i+j} \right) \in \mathbb{Z}_q[X]/(X^n + 1)$$

Since $X^n = -1$ then $X^{n+k} = -X^k$
 $\implies \mathcal{O}(n^2)$ naively, $\mathcal{O}(n \log n)$ using NTT.

Incomplete Timeline

- 1978 Knapsack cryptosystems attacked using lattice methods
- 1996 NTRU encryption
- 2005 Learning With Errors (LWE)
- 2008 Practical secure lattice-signature schemes
- 2009 Fully Homomorphic Encryption (FHE)
- 2016 NewHope (Ring-LWE KEM)
- 2017 Kyber (Module-LWE based) Dilithium (Module-LWE Module-SIS based)
- 2020s standardization: ML-KEM (Kyber) and ML-DSA (Dilithium)

Hard problems in lattices - SVP and CVP

- In elliptic curve cryptography, we rely on the hardness of the Discrete Logarithm problem:
given publicly known $g, p \in \mathbb{G}$, find privately known $x \in \mathbb{Z}$ such that $p = g^x$.
- In lattice based cryptography, we rely on problems related to finding short vectors in the lattice.

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Definition (SVP (Shortest Vector Problem))

find a shortest non-zero vector $x \in \mathcal{L}$, ie. that minimizes the Euclidean norm $\|x\|$, that is $\lambda_1(\mathcal{L}) = \|x\|$.

Intuition: find the shortest possible jump from the origin to another dot

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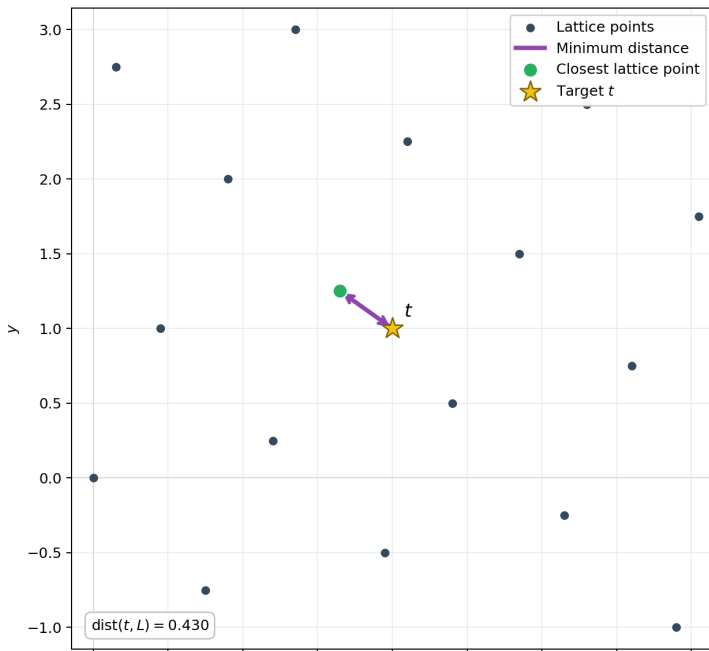
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Definition (CVP (Closest Vector Problem))

given a vector $y \in \mathbb{R}^n$ with $y \notin \mathcal{L}$, find a nonzero vector $x \in \mathcal{L}$ that is closest to y , ie. that minimizes the Euclidean norm $\|y - x\|$.

Intuition: find the grid dot that is closest to a given point (off the grid)

Closest Vector Problem (CVP)



Why Lattices?

- LLL algorithm is used to solve SVP and CVP.
 - LLL (Lenstra–Lenstra–Lovász): polynomial time method used for lattice basis reduction, which transforms a given basis of a lattice into a new basis that is shorter and nearly orthogonal, which makes SVP and CVP easier to approximate or solve using subsequent algorithms.

Intuition: make the lattice basis less skewed. Same grid of points, just a simpler basis. We don't change the lattice, we change the coordinate system used to describe it. → easier to solve SVP and CVP.

- improvement: BKZ reduction algorithm
- Both SVP and CVP become computationally difficult (NP hard) as the lattice dimension n grows (even on quantum computers)

NTRU

Definition (NTRU key recovery problem)

Given $h(x)$, find ternary polynomials $f, g \in \mathcal{T}$ satisfying

$$f(x) \cdot h(x) = g(x) \pmod{q} \in R_q$$

Where $R_q = \mathbb{Z}_q[X]/(X^n - 1)$.

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- 1998, was first scheme using polynomial rings, which latter where approached as algebraically structured lattices
 - "lattice-based" bcs recovering the small secret polynomials can be formulated and attacked as a lattice problem
 - earlier lattice usage: in 1978, Knapsack cryptosystems targeted by lattice attacks
- following the approach, in 2002, Micciano modified Ajtai's function to work over polynomial rings getting efficiency improvements

Hard problems in lattices - SIS

Definition (SIS (Short Integer Solution))

given $q \in \mathbb{Z}$ and matrix $A \in \mathbb{Z}_q^{m \times n}$, find a non-zero short $z \in \mathbb{Z}^n$ s.th.

$$A \cdot z = 0 \pmod{q}$$

"short": $z \in \{B, B\}^n$ for some bound B , often $z \in \{-1, 0, 1\}^n$.

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The set of all solutions to the SIS problem forms the (q -ary) lattice

$$\Lambda_q^\perp(A) = \{z \in \mathbb{Z}^n \mid A \cdot z = 0 \pmod{q}\}$$

where the SIS problem becomes the SVP for $\Lambda_q^\perp(A)$

Intuition: find a short nonzero vector in the lattice $\Lambda_q^\perp(A)$

Definition (Ring SIS (Ring Short Integer Solution))

given a uniformly random matrix $A \in {}^R R_q^n$, with elements $a_i \in R_q$,
find a non-zero integer vector $z \in R^n$ of norm $\|z\| \leq \beta$ such that

$$Az = \sum_i a_i \cdot z_i = 0 \in R_q$$

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Definition (MSIS (Module Short Integer Solution))

given a uniformly random matrix $A \in {}^R R_q^{m \times n}$, with columns
 $a_i \in R_q^m$, find a non-zero integer vector $z \in R^n$ of norm $\|z\| \leq \beta$
such that

$$Az = 0 \in R_q^m$$

$$\text{ie. } \sum_i a_{ij} \cdot z_j = 0 \in R_q, \text{ for } i = 1, \dots, m$$

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given a uniformly random matrix $A \in^R R_q^{m \times n}$, with columns $a_i \in R_q^m$, find a non-zero integer vector $z \in R^n$ of norm $\|z\| \leq \beta$ such that

$$Az = 0 \in R_q^m$$

ie. $\sum_i a_{ij} \cdot z_j = 0 \in R_q, \text{ for } i = 1, \dots, m$

Notice that Module-SIS is a Ring-SIS with $rank = 1$ ($m = 1$).

- without $\|z\| \leq \beta$, it is easy to find a solution via *Gaussian elimination*.
- $\beta < q$, because otherwise $z = (q, 0, \dots, 0) \in R^m$ would always be a valid solution.
- β and m must be large enough so that a solution is guaranteed to exist, usually $\beta \geq \sqrt{\hat{m}}$ and $m \geq \hat{m}$ where $\hat{m}$ is the smallest integer greater than $n \log q$.

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(example of decomposing values to keep small norms when doing operations in cryptographic protocols (eg. in LaBRADOR))

Hard problems in lattices

Definition (LWE (Learning With Errors))

choose a matrix $A \in {}^R \mathbb{Z}_q^{m \times n}$, a secret vector $s \in {}^R \mathbb{Z}_q^n$, and an error vector $e \in {}^R [-B, B]^m$. Let $b = As + e \pmod{q} \in \mathbb{Z}_q^m$. Given the pair (A, b) , the goal is to recover s .

Search-LWE & Decisional-LWE problems.

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For LWE samples, vector b is relatively close to exactly one vector in the LWE lattice:

$$\begin{aligned}\Lambda_q(A) &= \{A^T s \mid s \in \mathbb{Z}_q^n\} + q\mathbb{Z}^m \\ &= \{A^T s + qz \mid s \in \mathbb{Z}_q^n, z \in \mathbb{Z}^m\}\end{aligned}$$

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Observe: if $e = 0$, is a system of linear equations, which can be efficiently determined using Gaussian elimination over $\mathbb{Z}_q$. Hardness comes from the noise/error term $e \neq 0$ (and the norm bound). Similarly as with SIS, we also have Ring-LWE and Module-LWE variants of LWE.

Plain - Module - Ring

$$R_q = \mathbb{Z}_q[x]/(x^n + 1)$$

Assumption	Public linear relation	Underlying algebra
Plain LWE	$\mathbf{b} = A\mathbf{s} + \mathbf{e} \pmod{q}$	$A \in \mathbb{Z}_q^{m \times n}$
Module-LWE	$\mathbf{b} = A\mathbf{s} + \mathbf{e} \pmod{q}$	$A \in R_q^{k \times k}, \quad \mathbf{s}, \mathbf{e}, \mathbf{b} \in R_q^k$
Ring-LWE	$b = as + e \pmod{q}$	$a, s, e, b \in R_q$

Assumption	Public linear relation	Underlying algebra
Plain SIS	$A\mathbf{z} = \mathbf{0} \pmod{q}, \quad \mathbf{z} \neq \mathbf{0}, \quad \ \mathbf{z}\ \text{ small}$	$A \in \mathbb{Z}_q^{m \times n}, \quad \mathbf{z} \in \mathbb{Z}^n$
Module-SIS	$A\mathbf{z} = \mathbf{0} \pmod{q}, \quad \mathbf{z} \neq \mathbf{0}, \quad \ \mathbf{z}\ \text{ small}$	$A \in R_q^{m \times n}, \quad \mathbf{z} \in R^n$
Ring-SIS	$az = 0 \pmod{q}, \quad z \neq 0, \quad \ z\ \text{ small}$	$a \in R_q, \quad z \in R$

Dual lattices

SIS lattice:

$$\mathcal{L}^\perp(A) = \{z \in \mathbb{Z}^n \mid A \cdot z = 0 \pmod{q}\}$$

LWE lattice:

$$\begin{aligned}\mathcal{L}(A) &= \{A^T s \mid s \in \mathbb{Z}_q^n\} + q\mathbb{Z}^m \\ &= \{A^T s + qz \mid s \in \mathbb{Z}_q^n, z \in \mathbb{Z}^m\}\end{aligned}$$

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Dual of a lattice:

$$\mathcal{L}^* = \{y \in \mathbb{R}^m \mid \langle y, x \rangle \in \mathbb{Z} \forall x \in \mathcal{L}\}$$

ie. all vectors that measure lattice points with integers.

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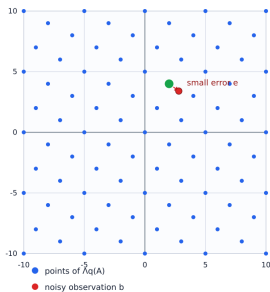
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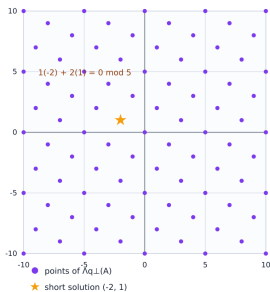
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SIS lattices and LWE lattices are dual to each other

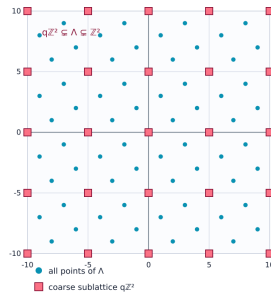
$$\mathcal{L}(A) = q\mathcal{L}^{\perp*}(A) \quad \text{ie.} \quad \frac{1}{q}\mathcal{L}(A) = \mathcal{L}^{\perp*}(A)$$

$q = 5$ Toy LWE: decode near $\Lambda q(A)$ 

$$\Lambda q(A) = \{s(1,2) + 5z : s \in \mathbb{Z}, z \in \mathbb{Z}^2\}$$

Toy SIS: $Az = 0 \pmod{q}$ 

$$\Lambda q_{\perp}(A) = \{z \in \mathbb{Z}^2 : ((1,2), z) = 0 \pmod{5}\}$$

q-ary lattice versus $q\mathbb{Z}^2$ 

$$\Lambda = 5\mathbb{Z}^2 + \mathbb{Z}(2,1) = \{5z + t(2,1) : z \in \mathbb{Z}^2, t \in \mathbb{Z}\}$$

Ajtai commitments

Definition (Ajtai commitment)

For a public random matrix $A \in \mathbb{Z}_q^{m \times n}$, embedding matrix $G \in \mathbb{Z}_q^{m \times l}$ and a short random vector $r \in \{-B, B\}^n$ (for bound B), an Ajtai commitment to the message $x \in \mathbb{Z}_q^l$ (bounded to B) is

$$c = Ar + Gx \pmod{q} \in \mathbb{Z}_q^n$$

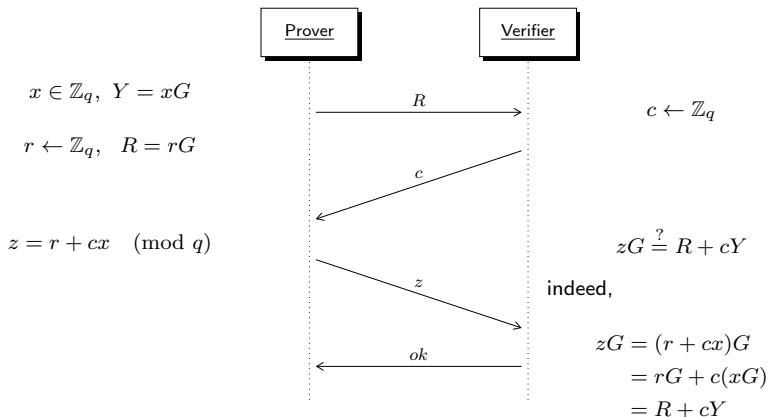
Observe that this is closely related to the SIS problem ($Az = 0 \pmod{q}$), and it relies on its hardness.

Lattice-based cryptography

- ML-KEM (Kyber): Key Encapsulation Mechanism. ie. Alice encapsulates a (symmetric) key using Bob's public key
- ML-DSA (Dilithium): signature scheme
- FN-DSA (Falcon): signature scheme, based on NTRU
- LaBRADOR: SNARK scheme
- FHE (BFV, BGV, CKKS, TFHE)

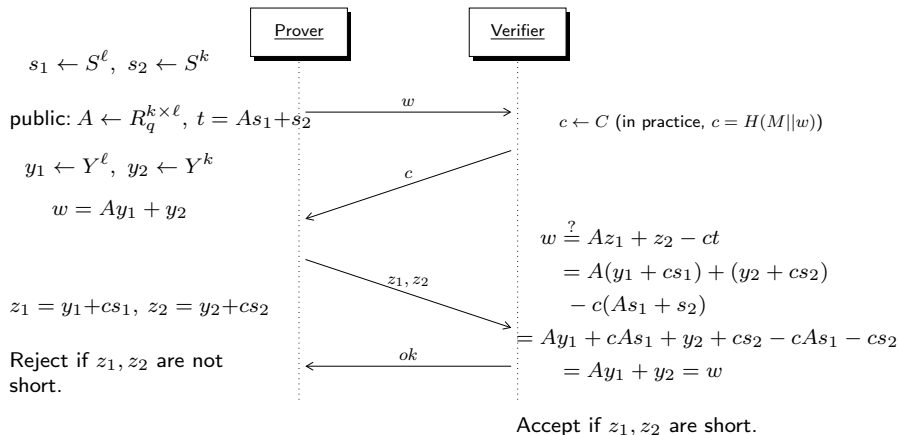
Recall: Sigma protocol over elliptic curve

$$G \in E(\mathbb{F}_p), \text{ ord}(G) = q, \text{ pub key } Y = xG$$



ML-DSA signatures

$$R_q = \mathbb{Z}_q[X]/(X^n + 1)$$



Falcon signatures

LaBRADOR SNARK

Benchmarks

Digital signatures

Scheme	Assumption	Public key size	Signature size
ECDSA-secp256k1	elliptic curves	33 B	64 B
Falcon-512	NTRU lattice	897 B	666 B
ML-DSA-44	Module-LWE / Module-SIS	1.31 KB	2.42 KB
XMSS-SHA2-10-256	hash based, stateful	64 B	2.50 KB
SLH-DSA-128s	hash based, stateless	32 B	7.86 KB

SNARKs / STARKs

Scheme	Assumption / technique	Proof size
Groth16	pairings / elliptic curves	128 B
LaBRADOR	Module-SIS	50–60 KB
Flock	binary fields + coding theory	200–400 KiB
Binius64	binary fields + sumcheck	200–500 KiB
Plonky2	Plonk + FRI	40–200 KB
Plonky3 / FRI-STARK	polynomial IOP + FRI	1–3 MiB

Conclusions

- not a single assumption: NTRU, SIS, R-SIS, M-SIS, LWE, R-LWE, M-LWE, ...